Foundations of Generative AI (ICT4403) - Assignment 2

Next-Word Prediction and Temperature-Controlled Text Generation: Simple RNN vs Bidirectional RNN vs LSTM vs GRU

Field	Details
Student Name	Unmesh
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Course	Foundations of Generative AI (ICT4403)
Dataset	A Study in Scarlet - Arthur Conan Doyle (Project Gutenberg #244)
Dataset Link	https://www.gutenberg.org/ebooks/244
GitHub Repository	https://github.com/unmesh1621-ai/fgai-rnn-text-generation

1. Objective

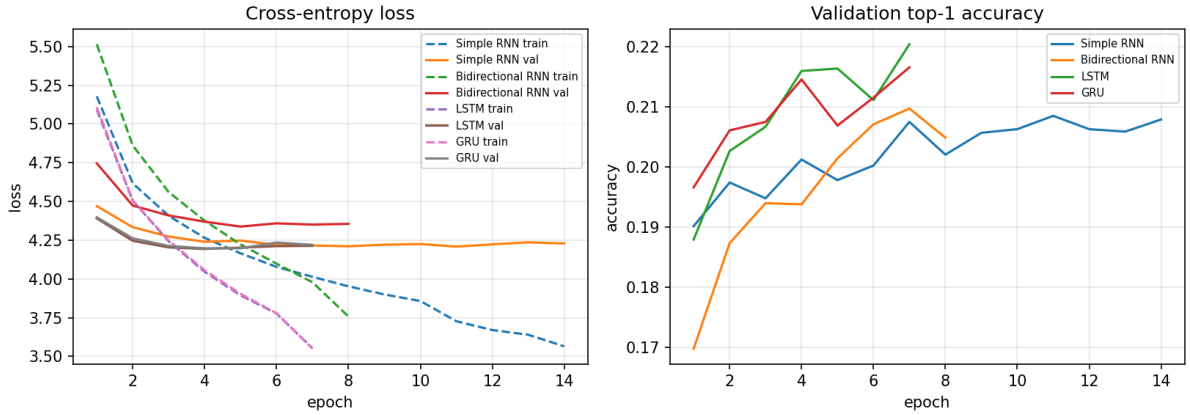
Train four word-level recurrent language models that predict the next word from the previous 10 words, compare them on held-out text, and study how sampling temperature changes the quality and diversity of generated text.

2. Data and Method

- Corpus: 49,681 tokens after removing the Gutenberg header/footer, lowercasing and splitting words and punctuation.
- Contiguous 80/10/10 split (39,744 / 4,968 / 4,969 tokens), so no window appears in two splits.
- Vocabulary built from the training split only (words seen at least 3 times): 1,543 entries; 12.2% of test tokens are unknown.
- Shared architecture: Embedding(256) -> recurrent core (256 units) -> LayerNorm -> Dropout(0.4) -> Linear(vocab). The BiRNN averages both directions over the window.
- Training: AdamW (lr 2e-3), batch 128, gradient clipping 0.5, learning-rate halving on plateau, early stopping (patience 3); the best validation checkpoint is tested.
- Hyperparameters were tuned on validation loss only; the test set was used once.

3. Test Results

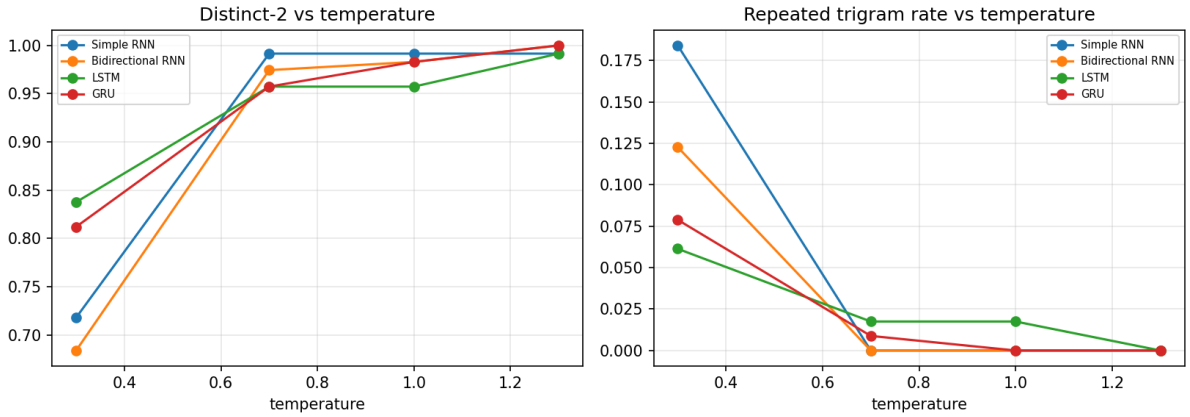
Model	Params	Best epoch	Loss	Perplexity	Top-1	Top-5
Simple RNN	923,655	11	4.120	61.6	21.3%	47.8%
Bidirectional RNN	1,450,759	5	4.285	72.6	19.7%	44.6%
LSTM	1,318,407	4	4.099	60.3	21.9%	47.2%
GRU	1,186,823	4	4.130	62.2	21.2%	47.0%



4. Temperature Analysis

Each model continued three prompts for 40 words at T = 0.3, 0.7, 1.0 and 1.3. Only newly generated words are scored. Distinct-n is unique n-grams divided by total n-grams; repeated trigrams is the share of trigrams already seen in the same output.

Model	T	Distinct-1	Distinct-2	Repeated trigrams
Simple RNN	0.3	0.533	0.718	0.184
Simple RNN	0.7	0.783	0.992	0.000
Simple RNN	1.0	0.783	0.992	0.000
Simple RNN	1.3	0.858	0.992	0.000
Bidirectional RNN	0.3	0.442	0.684	0.123
Bidirectional RNN	0.7	0.683	0.974	0.000
Bidirectional RNN	1.0	0.767	0.983	0.000
Bidirectional RNN	1.3	0.875	1.000	0.000
LSTM	0.3	0.600	0.838	0.061
LSTM	0.7	0.733	0.957	0.018
LSTM	1.0	0.750	0.957	0.018
LSTM	1.3	0.842	0.992	0.000
GRU	0.3	0.608	0.812	0.079
GRU	0.7	0.725	0.957	0.009
GRU	1.0	0.800	0.983	0.000
GRU	1.3	0.858	1.000	0.000



5. Conclusions

- LSTM is the best model, with test perplexity 60.3 and top-1 accuracy 21.9%. Gating helps keep useful context across the window.
- The Simple RNN came close to the gated models: on a short 10-word window the vanishing-gradient problem is mild.
- The Bidirectional RNN has the most parameters but the worst scores. Reading the window backwards adds little when the last few words matter most, and it overfits sooner. It stays autoregressive because its backward pass only reads past words in the window.
- Low temperature (0.3) gives locally fluent but looping text; 0.7 gives the best balance; 1.0-1.3 maximises diversity but breaks grammar. Diversity metrics must be read together with perplexity and the actual samples.